SEQUENCE AND SERIES ...

SUB-TOPICS

- definitions of Sequence, Series, Progression.
- AP
- GP
- HP.
- AGP.
- Misc Series.
- Properties of AP GP HP.
- Conditions when n AM's GM's HM's inserted b/w two numbers.
- Relation b/w AP GP HP.
- Method and when to use AM ≥ GM
- Sigma Notation.
- Methods of differences
- V_n Method.

FORMULAS

KEY CONCEPTS ...

Series:

: AP $\rightarrow 2b = a+c$

: GP $\rightarrow b^2 = ac$

: HP $\rightarrow \frac{2}{b} = \frac{1}{a} + \frac{1}{c}$

: AGP $\rightarrow$ form of Both AP, GP

: Special forms

*

$$(GM)^2 = (AM)(HM)$$

AP insertion:

d when n AM's inserted b/w a, b $d = \frac{b-a}{n+1}$

Sum of n AM's inserted b/w a, b $S = \frac{n}{2}(a+b)$

** $T_n = S_n - S_{n-1}$ ($\because$ for AP)

** If abc are in GP then $\frac{1}{a} \frac{1}{b} \frac{1}{c}$ are also in GP

*** log a, log b, log c are in AP if a, b, c is in GP.

For GP: difference $d = \frac{t_2}{t_1}$

$$S_\infty = \frac{a}{1-r}$$

$$S_n = a\left(\frac{r^n-1}{r-1}\right) \quad \because |r| > 1$$

$$= a\left(\frac{1-r^n}{1-r}\right) \quad \because |r| < 1$$

For AP: difference $d = t_2 - t_1$

$$S_n = \frac{n}{2}[2a + (n-1)d]$$

$$a_n = a + (n-1)d$$

For HP:

$$\boxed{t_n = \frac{ab}{b+(n-1)a-b}}$$

* $AM \geq GM \geq HP$ ($\because$ for $a_1 = a_2 = a_3 = \ldots = a_n$)

$AM = \frac{a+b}{2} \Rightarrow \frac{a_1+a_2+a_3+\ldots+a_n}{n}$

$GM = \sqrt{ab} \Rightarrow \sqrt[n]{a_1a_2a_3\ldots a_n}$

$HP = \frac{2ab}{a+b} \Rightarrow \frac{n}{\frac{1}{a_1}+\frac{1}{a_2}+\ldots+\frac{1}{a_n}}$

GP insertion

- r when n GMs inserted b/w a, b $r = \left(\frac{b}{a}\right)^{\frac{1}{n+1}}$
- Product of n GMs b/w ab is given by $(\sqrt{ab})^n$

HP insertion

d when n HMs inserted b/w ab $d = \frac{a-b}{ab(n+1)}$

Sigma Notation:

$\sum_1^n n = \frac{n(n+1)}{2}$

$\sum_1^n 1 = n$

$\sum n^2 = \frac{n(n+1)(2n+1)}{6}$

$\sum n^3 = \left[\frac{n(n+1)}{2}\right]^2$

*** $\boxed{\sum \frac{1}{n} \neq \frac{\sum 1}{\sum n}}$ & $\boxed{\sum n.n' \neq \sum n . \sum n'}$

: For decimal series take the No common and mult, Divide by 9 and taken simply....

Method of diff :

: first find diff b/w each Terms get $d_1 d_2 d_3 \ldots$

$d_1 \Rightarrow$ Same $T_n = an+b$ form Exisist . $T_n = an+b$

$d_2 \Rightarrow$ Same $T_n = an^2+bn+c$ form Exsist. $T_n = an^2+bn+c$

$d_3 \Rightarrow$ Same $T_n = an^3+bn^2+cn+d$ form Exsist.

$T_n = an^3+bn^2+cn+d$

V_n - Method :

: Express the n^{th} term of Given sequence as difference of Two consequtive terms for cancelation of Terms !

*** Express Num using denominator terms !

For AP $a_1+a_{100} = a_2+a_{99}$ $(a_1 a_2 a_3 \ldots a_{99} a_{100})$

For GP : $a_1 a_{100} = a_2 a_{99}$ $(a_1 a_2 a_3 \ldots a_{99} a_{100})$.

Method of Simplification :

- Put the whole. Sum as S and Substrack nS where n may be $\frac{1}{2}$, 10... (Any Value) According to the Question...

Use : $\boxed{AM \geq GM.}$

$$a = \frac{a}{2}+\frac{a}{2} = \frac{a}{3}+\frac{a}{3}+\frac{a}{3} = \frac{a}{4}+\frac{a}{4}+\frac{a}{4}+\frac{a}{4} \pm \frac{a}{5} + \cdots \frac{a}{5}.$$

idea.

- For Sum first get General term .. and use General formulas of Sigma Notations.
- In V_n Method : After Cancelletions if +Term in first is remained then −Term remains in last term...

Case ii: n is Odd : $n=2K-1$

$1^2-2^2+3^2-4^2+\dots\dots+(2K-1)^2-2K^2+2K^2$

$-K(2K+1)+4K^2$

$-2K^2-K+4K^2$

$2K^2-K$

$K(2K-1) \Rightarrow \frac{n+1}{2}(n) = \frac{n(n+1)}{2} = S_n$.

OPTION:

$(-1)^{n+1}\frac{n(n+1)}{2}$

Methods Of differences :

let

$S = 2+5+8+11+ \dots \bigcirc \rightarrow T_n$

$D_1 : 3,3,3 \dots$ ($\because$ first difference is containing all Same Values $T_n = an+b$ form.

$T_1 = a+b = 2$

$T_2 = 2a+b = 5$

$a=3$

$b=-1$.

$T_n = 3n-1$

Hence $S_n = \Sigma T_n$

$= \Sigma(3n-1)$

$= \Sigma 3n - \Sigma 1$

$= 3\Sigma n - \Sigma 1$

$= \frac{3n(n+1)}{2} - n$.

10. find the Sum Of $1+2+4+7+11+ \dots$ upto n terms

Sol:

$D_1 : 1\ 2\ 3\ 4$

$D_2 : 1\ 1\ 1$ $\boxed{T_n = an^2+bn+c}$

4Q. $77+757+7557+\ldots+7\overbrace{5\ldots5}^{98}7 = \dfrac{7\overbrace{5\ldots5}^{99}7+m}{n}$

$S = 77+757+7557+\ldots+75\ldots57$

$10S = 770+7570+75570+\ldots+75\ldots570$

$9S = -77+13+13+13+\ldots 98\text{times} + 75\ldots70$

$9S = -77+13\times98+75\ldots570$

$S = \dfrac{-77+1274+75\ldots57+13}{9}$

$S = \dfrac{1210+75\ldots57}{9}$

Hence: $m+n = 1210+9$

$= 1219.$

5Q. $\displaystyle\sum_{K=1}^{\infty}\sum_{n=1}^{\infty}\frac{K}{2^{n+K}} = 2$

Solution:

$S = \displaystyle\sum_{K=1}^{\infty}\sum_{n=1}^{\infty}\frac{K}{2^{n+K}}$

$S = \displaystyle\sum_{K=1}^{\infty}\sum_{n=1}^{\infty}\frac{K}{2^K}\cdot\frac{1}{2^n}$

$S = \displaystyle\sum_{K=1}^{\infty}\frac{K}{2^K}\left(\sum_{n=1}^{\infty}\frac{1}{2^n}\right)$

$S = \displaystyle\sum_{K=1}^{\infty}\frac{K}{2^K}\left(\frac{1}{2}+\frac{1}{2^2}+\frac{1}{2^3}+\ldots\right)$

$S = \displaystyle\sum_{K=1}^{\infty}\frac{K}{2^K}\left(\frac{\cancel{1/2}^{\,1}}{\cancel{1/2}}\right)$

$S = \displaystyle\sum_{K=1}^{\infty}\frac{K}{2^K} = \left(\frac{1}{2^1}+\frac{2}{2^2}+\frac{3}{2^3}+\ldots\right)$ — ①

$\dfrac{S}{2} = \dfrac{1}{2^2}+\dfrac{2}{2^3}+\dfrac{3}{2^3}$ — ②

eq ② - ①

$$\frac{S}{2} = \frac{1}{2} + \frac{1}{2} + \frac{1}{2^3} + \dots = 1$$

$$S = 2$$

Q. $\sum_{r=1}^{10} (r^2+1) \times r!$

T_r (General Term) $= (r^2+1) \times r!$

$T_r = [r(r+1) - (r-1)] \times r!$

$T_r = r(r+1)! - (r-1)r!$

This is remaining...

$T_1 = \cancel{1(2)!} - (0)$

$T_2 = \cancel{2(3)!} - \cancel{1(2!)}$

$T_3 = \cancel{3(4)!} - \cancel{2(3!)}$

$T_4 = 4(5)! - \cancel{(3)(4!)}$

$\vdots$

$T_{10} = 10(11)! - (9)10!$

$= 10(11 \times 10!) - 9 \cdot 10!$

$= 10!(101)$

$S_n = (n+1)!\,n - 0.$

$S_{10} = 11!\,10.$

Q.

1

2 3

4 5 6

7 8 9 10

$1 + [2+3] + [4+5+6] + \dots$

1 2 3 ...

$a + (n-1)d$

$1 + (n-1) \Rightarrow n = a_n$

$k^{th} \rightarrow$ Have K Numbers

$\rightarrow$ Which row No. 5310 is there..?

Ans: 103

$1 + 2 + 4 + 7 + \dots + T_n.$